

What the world's worst outbreak can teach its neighbours

A COVID-19 data story

Luis Faria · BDA601 Big Data and Analytics · Assessment 3

Johns Hopkins CSSE confirmed cases · 22 Jan 2020 - 9 Mar 2023

The US recorded **103 million** cases. The question that matters for its neighbours:
could they have seen their own waves coming?

The approach

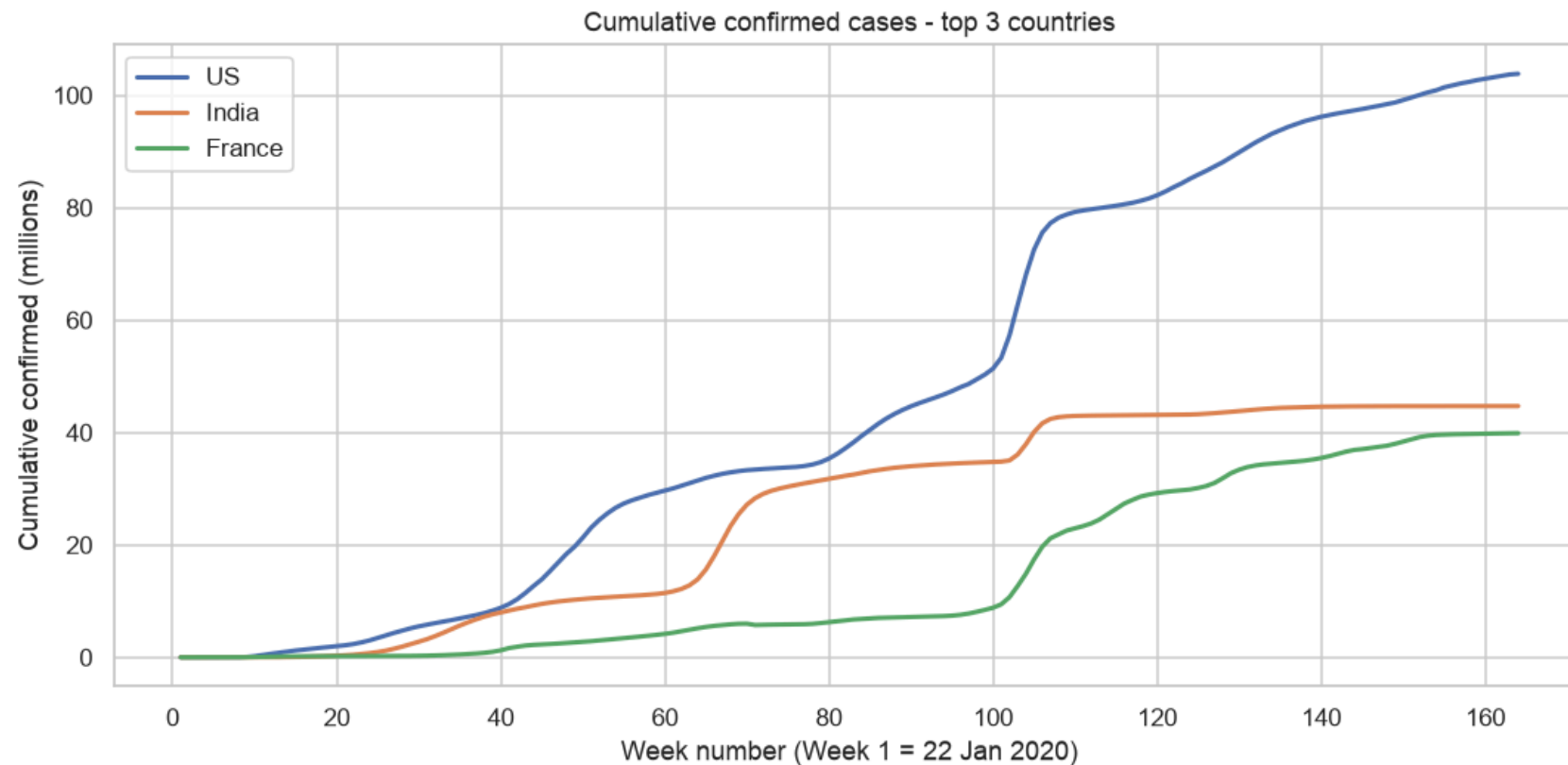
Every big-data project runs **prepare → analyse → decide**.

One analytical chain, each step feeding the next:

**top-3 countries → linear regression → pick the most volatile → K-Means waves → graph to neighbours
→ lead-lag → recommendation**

Tools: **Apache Spark MLlib** (regression + K-Means) · **networkx** (graph) · 164 weeks of data.

The three worst-hit countries

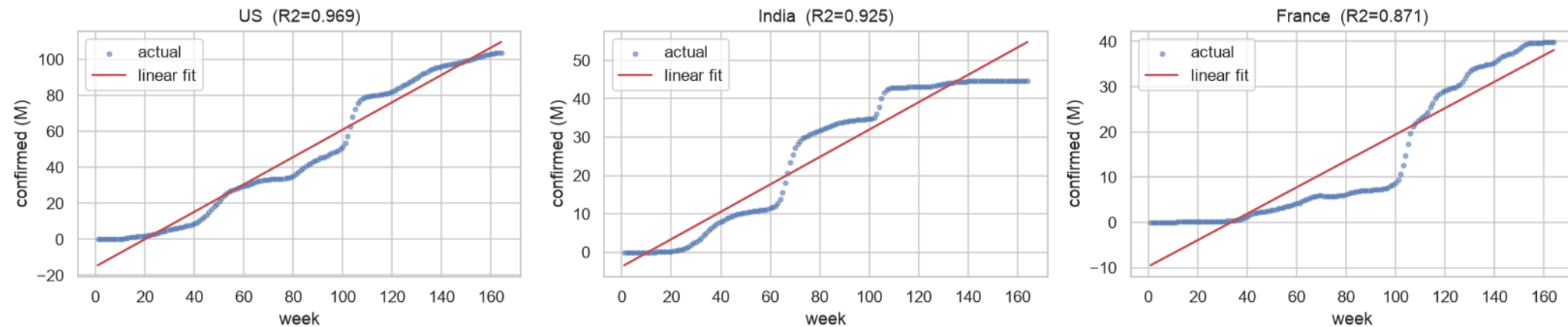


Top 3 by total confirmed: **US 103.8M · India 44.7M · France 39.9M.**

All three rise - but at very different rates and shapes. Which one is the *most volatile*?

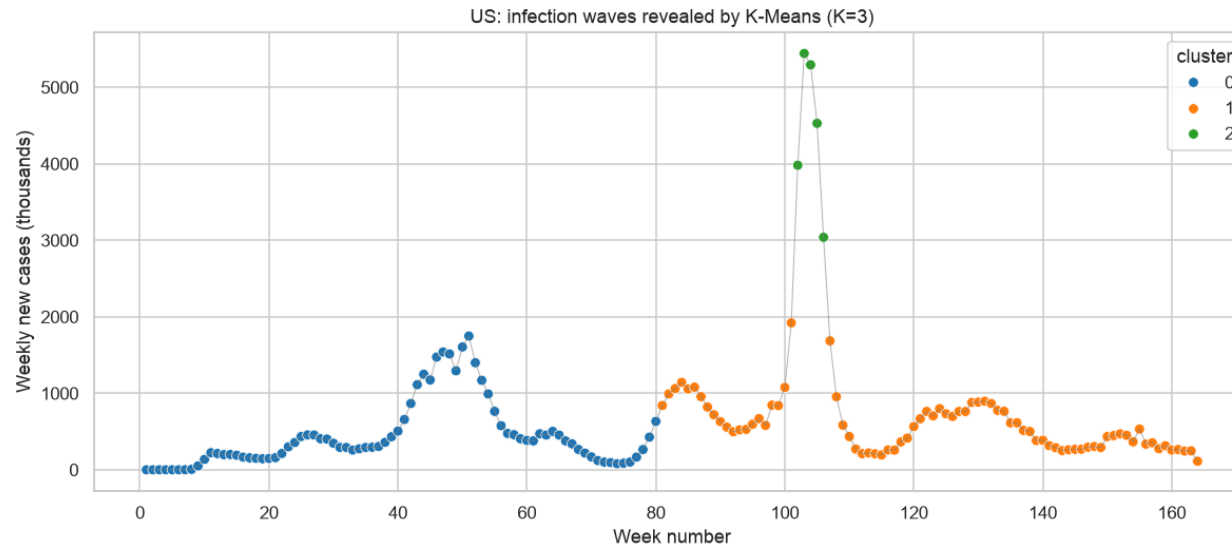
Predictive modelling: a line is not enough

Linear regression fits - cumulative confirmed vs week



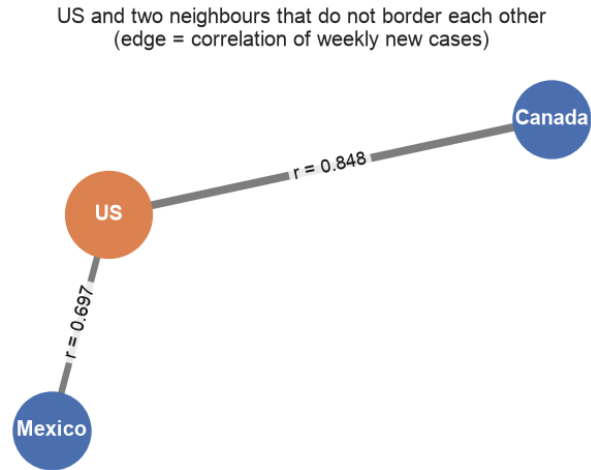
- Linear regression of cumulative cases on week number, per country.
- **US is the most volatile** - by the variance of its *weekly new cases* ($6.4e11$ vs India $2.5e11$, France $1.6e11$), not merely its size. Slope $\sim 760k$ cases/week, $R^2 = 0.97$.
- Key point: even $R^2 = 0.97$ **hides** the story - a straight line cannot show *when* the surges hit.

Clustering reveals the waves



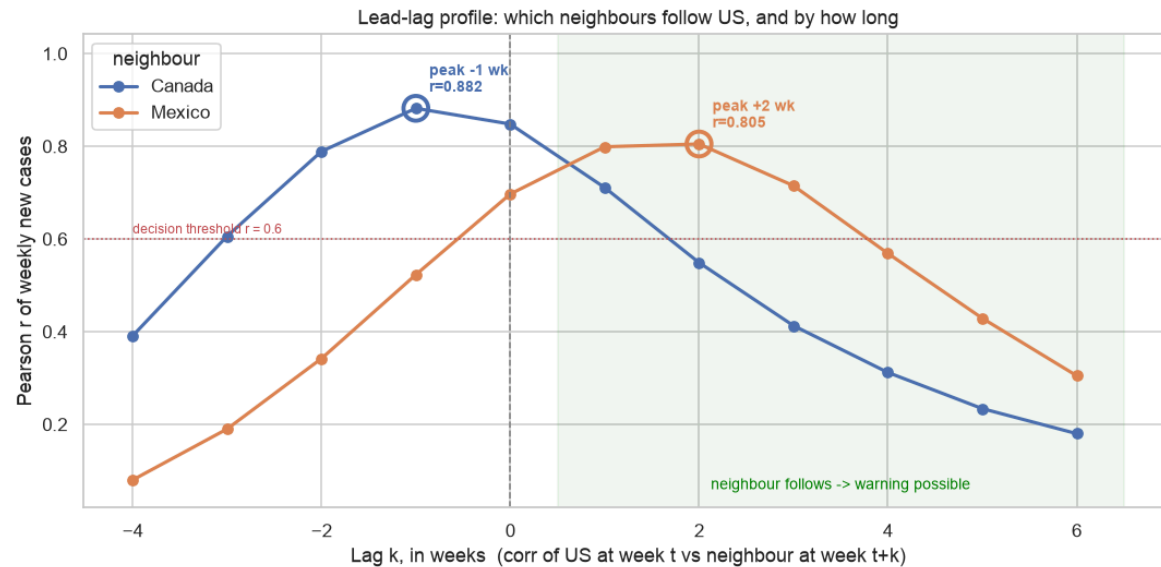
- K-Means on [week, weekly new cases] , best $K = 3$ (silhouette 0.705).
- It isolates a **mega-surge of ~4.46M new cases/week around weeks 102-106** (the Omicron wave, Jan 2022) as its own cluster.
- This **shows** growth was not steady: it came in waves (up - down - up).

Graph analytics: who moves with the US?



- US linked to **Canada** and **Mexico**, two neighbours that do not border *each other*.
- Edge = correlation of weekly new cases **in the same week**: Canada $r = 0.85$, Mexico $r = 0.70$.
- Obvious reading: Canada tracks the US most closely, so warn Canada.
- **But that is the wrong question.** It asks *do these curves look alike?*, not *does my wave arrive after theirs?*

Lead-lag: who can actually be warned?

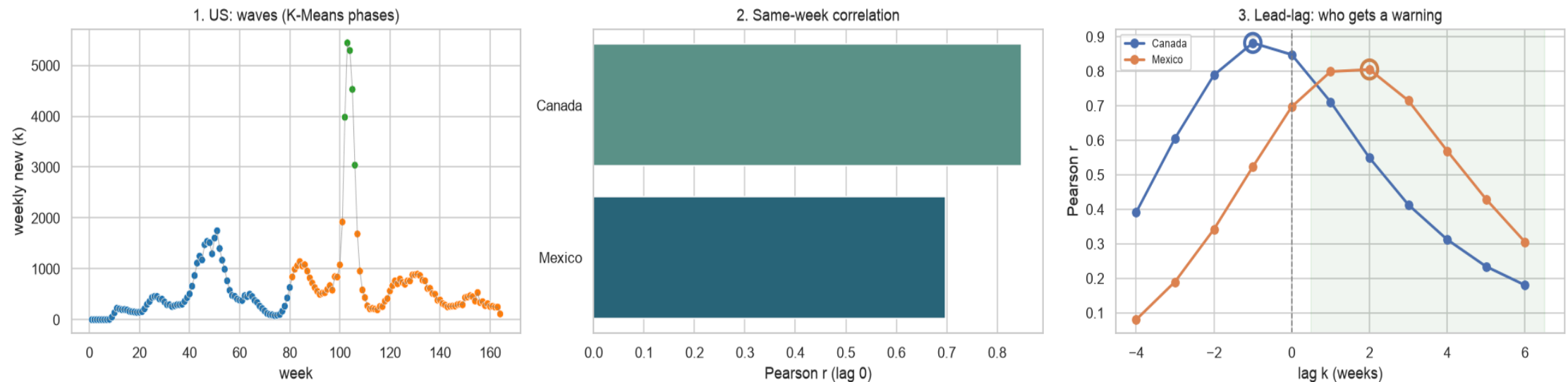


Correlation of US at week t vs neighbour at week $t+k$. **Rule set in advance:** warn only if peak lag $\geq +1$ week and $r \geq 0.60$.

- **Canada peaks at $k = -1$ ($r = 0.88$):** moves *with or ahead of* the US → no warning window.
- **Mexico peaks at $k = +2$ ($r = 0.81$, up from 0.70):** follows the US → ~2 weeks of warning.

The whole story in one frame

From raw cases to a neighbour early-warning recommendation



Left: US waves by phase. Middle: same-week correlation. Right: the lead-lag profile that overturns it.
Raw data → phases → *who actually gets a warning*.

Recommendations to the neighbours

- **Mexico (lag +2, $r = 0.81$):** the real early-warning case. When the US enters a surge phase, pre-position testing and hospital capacity **~2 weeks ahead**. Same-week correlation understated this.
- **Canada (lag -1, $r = 0.88$):** highly correlated but **synchronous** - the US is *not* a leading indicator here. Invest in domestic surveillance; by the time US numbers climb, Canada's already are.
- **General:** plan capacity for the isolated **Omicron-style mega-surge cluster**, not the steady baseline - that single cluster carried ~7x the overall weekly average.

Limitations & close

- **Data:** counts are *confirmed cases*, so they track testing and reporting as much as transmission; this file ends **9 Mar 2023**, and JHU ceased collection on **10 Mar 2023**.
- **Method:** `week` is a clustering input, so phases are partly temporal by construction; "neighbour" is geography, not mobility; one fixed lag averages over three years of variants.
- **Correlation is not causation** - a lagged match may reflect a shared driver, such as a variant reaching the region.
- **Next steps:** mobility and vaccination data, and a time-varying lag per wave.

Close: the same-week number said *Canada*. Shifting the series by two weeks said *Mexico* - and that is the recommendation that would actually have bought someone time.